

Ex. 3.3 | Ch. 3 | P 82 | Chloe | 2025.7.22

证: 令 $h = f + g$, 已知 $\langle h | \hat{Q} h \rangle = \langle \hat{Q} h | h \rangle$,

$$\text{则 } \langle f + g | \hat{Q} (f + g) \rangle = \langle \hat{Q} (f + g) | f + g \rangle$$

$$\Rightarrow \text{上式左边} = \langle f | \hat{Q} f \rangle + \langle f | \hat{Q} g \rangle + \langle g | \hat{Q} f \rangle + \langle g | \hat{Q} g \rangle$$

$$\text{上式右边} = \langle \hat{Q} f | f \rangle + \langle \hat{Q} f | g \rangle + \langle \hat{Q} g | f \rangle + \langle \hat{Q} g | g \rangle$$

$$\text{又 } \langle f | \hat{Q} f \rangle = \langle \hat{Q} f | f \rangle, \quad \langle g | \hat{Q} g \rangle = \langle \hat{Q} g | g \rangle$$

$$\text{故 } \langle f | \hat{Q} g \rangle + \langle g | \hat{Q} f \rangle = \langle \hat{Q} f | g \rangle + \langle \hat{Q} g | f \rangle \quad (1)$$

$$\text{令 } h = f + ig, \text{ 则 } \langle f + ig | \hat{Q} (f + ig) \rangle = \langle \hat{Q} (f + ig) | f + ig \rangle$$

$$\Rightarrow \text{上式左边} = \langle f | \hat{Q} f \rangle + \langle f | \hat{Q} (ig) \rangle + \langle ig | \hat{Q} f \rangle + \langle ig | \hat{Q} (ig) \rangle$$

$$= \langle f | \hat{Q} f \rangle + i \langle f | \hat{Q} g \rangle - i \langle g | \hat{Q} f \rangle + \langle g | \hat{Q} g \rangle$$

$$\text{上式右边} = \langle \hat{Q} f | f \rangle + \langle \hat{Q} f | ig \rangle + \langle \hat{Q} (ig) | f \rangle + \langle \hat{Q} (ig) | ig \rangle$$

$$= \langle \hat{Q} f | f \rangle + i \langle \hat{Q} f | g \rangle - i \langle \hat{Q} g | f \rangle + \langle \hat{Q} g | g \rangle \quad (2)$$

$$\text{故 } i \langle f | \hat{Q} g \rangle - i \langle g | \hat{Q} f \rangle = i \langle \hat{Q} f | g \rangle - i \langle \hat{Q} g | f \rangle$$

$$\Rightarrow \langle f | \hat{Q} g \rangle - \langle g | \hat{Q} f \rangle = \langle \hat{Q} f | g \rangle - \langle \hat{Q} g | f \rangle \quad (2)$$

$$(1) + (2) \text{ 得 } 2 \langle f | \hat{Q} g \rangle = 2 \langle \hat{Q} f | g \rangle$$

$$\text{故 } \langle f | \hat{Q} g \rangle = \langle \hat{Q} f | g \rangle \text{ 得证}$$